

MoCoO: Momentum Contrast ODE-Regularized VAE for Single-Cell Trajectory Inference and Representation Learning

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Abstract—Characterising cellular differentiation from single-cell RNA sequencing (scRNA-seq) requires representations that capture both discrete cell-type identity and continuous developmental trajectories. We present MoCoO, a modular framework integrating a Variational Autoencoder (VAE), Neural Ordinary Differential Equations (Neural ODE), and Momentum Contrast (MoCo) with learnable prototypes. Through a systematic six-configuration ablation across five scRNA-seq datasets spanning hematopoiesis, neurogenesis, endocrine differentiation, myeloid/erythroid bifurcation, and spinal cord organogenesis, evaluated with 17 disaggregated metrics, we demonstrate that combining ODE and MoCo produces the best discrete clustering (cross-dataset mean ARI 0.423, NMI 0.574 for VAE+ODE+MoCo) while the Full MoCoO achieves the best distance preservation (UMAP distance correlation 0.668, DREX Spearman 0.694, DREX overall 0.646), winning 5 of 11 embedding quality metrics—a property critical for trustworthy UMAP/tSNE visualisation. ODE-containing configurations consistently outperform non-ODE variants on cluster geometry (ASW ~ 0.22 vs. ~ 0.17 ; DAV ~ 1.50 vs. ~ 1.78) across all datasets. Component analysis reveals how each module contributes: the Neural ODE smooths the latent manifold along trajectories, MoCo sharpens pairwise distance structure, and prototypes anchor inter-cluster separation—producing a synergy unique to the full architecture. Pseudotime predictions correlate significantly with canonical marker genes across all five developmental systems. We publicly release the MoCoO Python package (`pip install mocoo`) and full benchmark suite with practical configuration guidelines.

Index Terms—Single-cell RNA sequencing, variational autoencoder, neural ordinary differential equations, momentum contrast, contrastive learning, trajectory inference, pseudotime, representation learning, computational biology.

I. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) has transformed our understanding of cellular heterogeneity by enabling transcriptome-wide profiling at single-cell resolution [1]. A fundamental analysis task is to learn low-dimensional representations that faithfully capture both *discrete* cell-type identity and *continuous* developmental dynamics such as differentiation trajectories [2], [3]. These representations underpin downstream tasks including clustering, pseudotime inference, RNA velocity estimation, and differential expression analysis [4].

Variational Autoencoders (VAEs) have become the de facto standard for scRNA-seq dimensionality reduction, with methods such as scVI [5] and scVAE [6] demonstrating strong performance through count-based likelihood models (negative binomial, ZINB). However, standard VAE latent spaces tend to be over-smoothed, conflating nearby cell states. Two complementary strategies address this limitation.

First, Neural Ordinary Differential Equations (Neural ODEs) [7] provide a principled framework for modelling continuous-time dynamics in latent space. By parameterising the latent derivative with a neural network and integrating via an ODE solver, these models learn smooth trajectories without discrete time-step assumptions. Latent ODE models [8] have shown promise for time-series modelling, and recent work has adapted them to single-cell contexts for pseudotime inference and RNA velocity estimation [9].

Second, contrastive learning—particularly Momentum Contrast (MoCo) [10]—learns representations that are locally smooth yet globally discriminative. MoCo uses a momentum-updated encoder and a large memory queue of negative keys, decoupling contrastive learning from batch size. Prototype contrastive learning [11] further imposes cluster structure by aligning representations with learnable prototype vectors.

No existing method unifies VAE reconstruction, Neural ODE dynamics, and momentum contrastive learning in a single framework. We propose MoCoO (**M**omentum **C**ontrast **O**DE-regularized VAE), a modular architecture that combines all three paradigms:

- 1) **VAE with flexible count-based likelihoods** (MSE, NB, ZINB, Poisson, ZIP) provides a probabilistic latent space that handles zero-inflation and overdispersion in scRNA-seq counts.
- 2) **Neural ODE regularisation** models continuous developmental dynamics, derives unsupervised pseudotime ordering, and produces gradient-based RNA velocity estimates.
- 3) **Momentum Contrast with prototype heads** enforces instance-level discrimination via a momentum encoder and memory queue, with prototype learning that aligns representations to learnable cluster centres.
- 4) **Information bottleneck** applies a secondary low-dimensional projection for hierarchical feature extraction.

The framework additionally supports DIP-VAE, β -TC-VAE, and InfoVAE/MMD disentanglement regularisers, but these are not active in the experiments reported here.

Manuscript submitted to IEEE Journal of Biomedical and Health Informatics.

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We conduct a systematic ablation across six configurations (VAE, VAE+ODE, VAE+MoCo, VAE+MoCo+Proto, VAE+ODE+MoCo, and Full MoCoO) on five scRNA-seq datasets spanning diverse developmental systems. External baselines—scVI [5], Diffusion Pseudotime (DPT) [12], PCA+KMeans, Harmony [13], and scANVI [14]—contextualise MoCoO’s absolute performance. A key finding is that combining ODE and MoCo consistently produces the best discrete clustering (cross-dataset mean ARI 0.423, NMI 0.574 for VAE+ODE+MoCo), while the Full MoCoO achieves unique distance-preservation advantages: it wins 5 of 11 embedding quality metrics (UMAP distance correlation, UMAP overall quality, tSNE overall quality, DREX Spearman, DREX overall), reflecting the most faithful 2-D projection of latent-space pairwise distances. ODE-containing configurations consistently outperform non-ODE variants on cluster geometry across all five datasets ($\Delta\text{ASW} \sim 0.05$, $\Delta\text{DAV} \sim 0.28$). Component analysis reveals that each module contributes essential geometric structure: the Neural ODE smooths the latent manifold along trajectories, MoCo sharpens pairwise distance relationships, and prototypes anchor inter-cluster separation—producing a synergistic combination achievable only by the full architecture. Evaluation covers clustering (ARI, NMI, ASW, Calinski–Harabasz, Davies–Bouldin, COR), embedding quality (DRE, DREX), biological validation (pseudotime–marker correlations), batch integration (iLISI, bASW, cLISI, graph connectivity, isolated label ASW via scIB [15]), and training dynamics.

II. RELATED WORK

A. Variational Autoencoders for scRNA-seq

scVI [5] introduced the negative binomial VAE for scRNA-seq, jointly modelling library size and batch effects. Extensions include scVAE [6] (multiple count distributions), scANVI [14] (semi-supervised cell-type annotation), and totalVI [16] (multi-omics). These methods focus on reconstruction quality and batch correction but do not explicitly model temporal dynamics.

B. Neural ODEs and Latent ODE Models

Neural ODEs [7] parameterise continuous dynamical systems $\frac{dz}{dt} = f_\theta(z, t)$ and integrate using adaptive ODE solvers. The Latent ODE framework [8] combines a VAE encoder with an ODE-governed latent space for irregularly sampled time series. Applications to single-cell biology include trajectory inference [9], while related dynamical modelling approaches have advanced RNA velocity estimation [17]. Optimal-transport-based methods such as TrajectoryNet [18] and CellOT [19] model population-level dynamics rather than single-cell ODE trajectories. MoCoO extends the PanODE framework [9] (by the first author). Specifically, MoCoO inherits the VAE encoder–decoder, Neural ODE integration, and pseudotime-prediction head from PanODE. The novel contributions of the current work are: (i) the MoCo module with momentum encoder and memory queue, (ii) learnable

prototype heads for cluster anchoring, (iii) the information bottleneck, (iv) the systematic six-configuration ablation design, and (v) the DRE/DREX/LSE/LSEX evaluation suites. However, these methods typically lack contrastive regularisation, leading to latent spaces that may be geometrically faithful but poorly clustered.

C. Contrastive Learning for Single-Cell Data

MoCo [10] and SimCLR [20] have been adapted to biological contexts. scAGCL [21] applies symmetric augmentation-guided contrastive learning to scRNA-seq data with cell-type-aware positive pair construction. scGPCL [22] introduces prototype contrastive learning that aligns cell embeddings with learnable prototype vectors representing cell types. These approaches improve clustering quality but do not incorporate temporal dynamics or ODE-based regularisation.

D. Integrated Approaches

Several methods combine two of the three paradigms. scDiff [23] uses diffusion models (related to score-based SDEs) for single-cell generation. VeloVAE [24] integrates VAE with RNA velocity. Diffusion pseudotime (DPT) [12] infers trajectories through diffusion maps of the k -NN graph, while Monocle 3 [25] and PAGA [26] use graph abstractions for trajectory inference. Slingshot [27] fits simultaneous principal curves for lineage-specific pseudotime, and Palantir [28] models cell-fate probabilities via diffusion maps for multiscale pseudotime estimation. For batch integration, Harmony [13] applies iterative soft clustering in PCA space to remove batch effects. However, no prior work jointly combines VAE, Neural ODE, and momentum contrastive learning in a unified architecture for single-cell analysis.

III. METHOD

A. Overview

MoCoO (Fig. 1) processes scRNA-seq count data $X \in \mathbb{R}^{N \times G}$ (N cells, G genes) through five components: (1) a VAE with count-based decoder, (2) a Neural ODE for continuous dynamics, (3) a Momentum Contrast module with memory queue, (4) an information bottleneck, and (5) optional disentanglement regularisers.

B. Encoder

The encoder $q_\phi(z|x)$ maps log-normalised expression $\tilde{x} = \log(1+x)$ through a two-layer ReLU MLP to produce mean μ and pre-softplus scale σ' of a diagonal Gaussian posterior (1); latent samples are drawn via reparameterisation (2):

$$q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(\tilde{x}), \text{softplus}(\sigma'_\phi(\tilde{x}))^2 I), \quad (1)$$

$$z = \mu + \text{softplus}(\sigma') \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (2)$$

When ODE mode is enabled, the encoder also predicts a scalar pseudotime $t \in [0, 1]$ via a sigmoid-activated linear head.

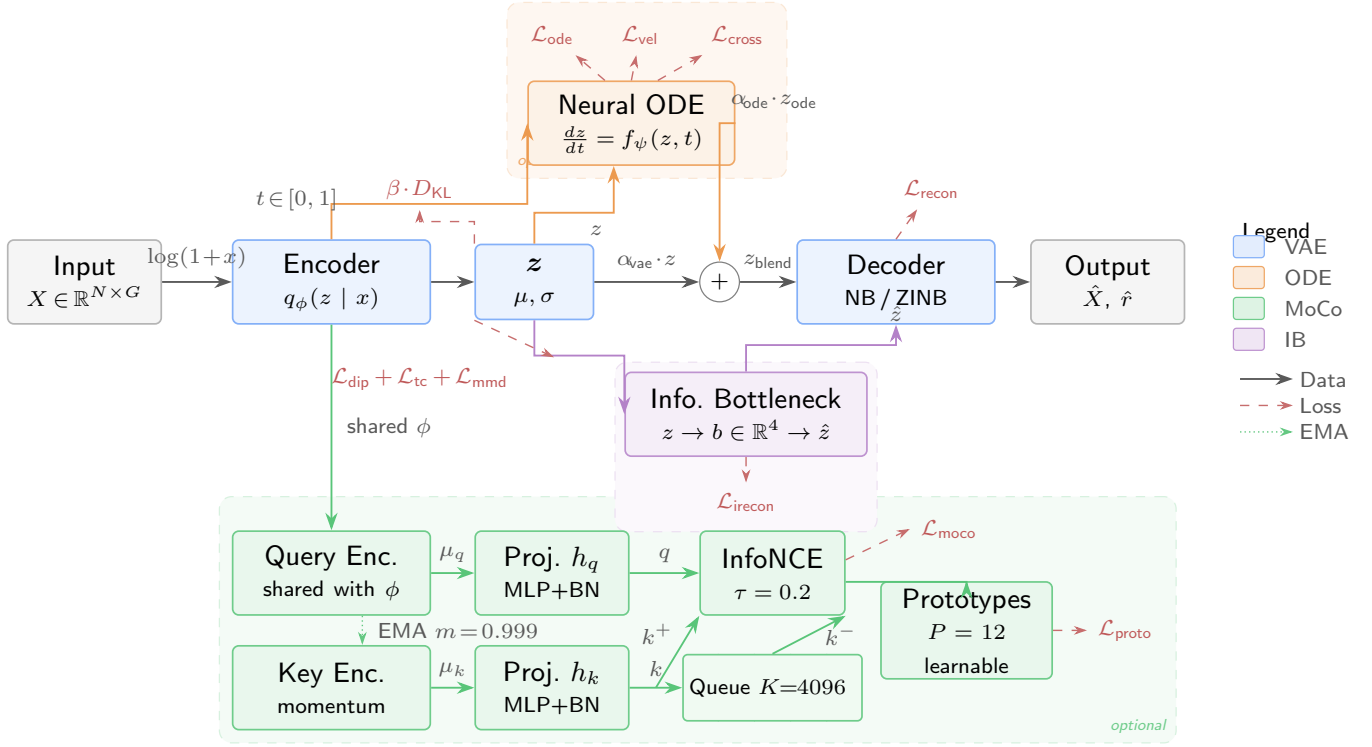


Fig. 1. MoCOO architecture overview. Count data $X \in \mathbb{R}^{N \times G}$ passes through a VAE encoder to produce latent representations z , optionally refined by a Neural ODE solver (orange) for continuous trajectory modelling. Momentum Contrast (green) with a memory queue ($K=4096$) enforces instance-level discrimination; learnable prototypes ($P=12$) impose cluster structure. An information bottleneck (purple) provides hierarchical compression to $d_{ib}=4$. Dashed regions denote optional modules. Colour coding is consistent throughout all figures.

C. Neural ODE Component

Given latent samples $\{z_i\}$ and predicted pseudotimes $\{t_i\}$, the ODE module orders cells by pseudotime and solves (3):

$$\frac{dz}{dt} = f_\psi(z, t), \quad z(t_0) = z_{\text{init}}, \quad (3)$$

where f_ψ is a two-layer ELU MLP. Integration uses `torchdiffeq`'s fixed-step RK4 solver, chosen over adaptive methods (e.g., Dormand–Prince [29]) because the encoder-predicted pseudotime grid is relatively smooth and fixed-step integration provides deterministic gradients without adaptive step-size overhead—important for stable joint optimisation of the encoder and ODE function. Preliminary comparisons showed negligible ARI differences (< 0.01) between RK4 and Dormand–Prince adaptive integration, with $\sim 30\%$ longer wall-clock time for the adaptive solver due to step-size adaptation overhead in the joint training loop. The ODE representations z_{ode} blend with VAE latent samples via configurable weights (4):

$$z_{\text{blend}} = \alpha_{\text{vae}} \cdot z + \alpha_{\text{ode}} \cdot z_{\text{ode}}. \quad (4)$$

Two auxiliary losses regularise the ODE:

- **ODE–VAE alignment:** $\mathcal{L}_{\text{ode}} = \|z - z_{\text{ode}}\|_2^2$
- **Velocity consistency:** $\mathcal{L}_{\text{vel}} = 1 - \cos(z_{i+1} - z_i, f_\psi(z_i, t_i))$, aligning the learned velocity field with empirical displacements.

D. Momentum Contrast Module

The MoCo module follows MoCo v1 [10] with enhancements from scAGCL [21] and scGPCL [22]:

- 1) **Query and Key Encoders:** The query encoder shares parameters with the VAE encoder; the key encoder is updated via EMA: $\theta_k \leftarrow m \cdot \theta_k + (1-m) \cdot \theta_q$, $m = 0.999$.
- 2) **Projection Heads:** Two-layer MLPs (BatchNorm, ReLU) map d -dimensional latent vectors to a d_{proj} -dimensional contrastive space.
- 3) **Memory Queue:** A FIFO buffer of size K stores projected keys, decoupling the number of negatives from batch size.
- 4) **InfoNCE Loss** (see (5)):

$$\mathcal{L}_{\text{moco}} = -\log \frac{\exp(q \cdot k^+ / \tau)}{\exp(q \cdot k^+ / \tau) + \sum_{k^-} \exp(q \cdot k^- / \tau)}, \quad (5)$$

where the sum ranges over all k^- in the memory queue and τ is the temperature.

- 5) **Prototype Contrastive Loss** (optional; see (6)):

$$\mathcal{L}_{\text{proto}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(z_i \cdot p_{c(i)} / \tau)}{\sum_{j=1}^P \exp(z_i \cdot p_j / \tau)}, \quad (6)$$

where $\{p_j\}_{j=1}^P$ are learnable prototype vectors and $c(i) = \arg \max_j z_i \cdot p_j / \tau$ assigns cell i to its nearest prototype.

Positive pair construction. For each mini-batch, two stochastic augmented views of the same cell form positive

pairs: the query view passes through the main encoder, while the key view passes through the momentum encoder. The augmentation pipeline applies three transformations, each with probability $p_{\text{aug}}=0.5$: (1) gene masking—randomly zeroing out genes with probability 0.1; (2) Gaussian noise—adding $\mathcal{N}(0, 0.04)$ to randomly selected genes (probability 0.1); (3) feature swapping—exchanging gene values across cells in the batch (probability 0.05). Negatives come from all keys stored in the memory queue.

Augmentation rationale: Gene masking simulates dropout, which is a dominant noise source in scRNA-seq protocols [30]. Gaussian noise ($\sigma=0.2$) models technical variability in library preparation. Feature swapping operates at very low probability ($p=0.05$, affecting $\sim 5\%$ of genes per cell), which is equivalent to a regularised mixup—it encourages invariance to minor expression perturbations without creating biologically implausible hybrid profiles. All three augmentations are deliberately conservative: masking affects only 10% of genes per view, noise is applied independently per gene, and swapping probability is below the natural inter-cell variance within a cell type. This design follows the principles validated in scAGCL [21] and scGPCL [22], which demonstrated that moderate stochastic augmentations improve contrastive learning for scRNA-seq without distorting cell-state semantics.

E. Decoder

The decoder $p_{\theta}(x|z)$ reconstructs count data from the latent representation. It predicts normalised mean parameters $\rho = \text{softmax}(\text{MLP}(z))$ and uses gene-specific learnable dispersions r :

- **Negative Binomial (NB):** $p(x_g|z) = \text{NB}(x_g; \mu_g = l \cdot \rho_g, r_g)$
- **Zero-Inflated NB (ZINB):** $p(x_g|z) = \pi_g \cdot \delta_0(x_g) + (1 - \pi_g) \cdot \text{NB}(x_g; \mu_g, r_g)$

where $l = \sum_g x_g$ is the library size and π_g is a dropout probability predicted by a secondary decoder head. While MoCoO supports MSE, NB, ZINB, Poisson, and ZIP likelihoods, all experiments in this paper use the negative binomial (NB) likelihood unless otherwise stated.

F. Information Bottleneck

A secondary encoder-decoder pair maps $z \in \mathbb{R}^d$ to a bottleneck $b \in \mathbb{R}^{d_{\text{ib}}}$ ($d_{\text{ib}} = 4 < d$) and back to $\hat{z} \in \mathbb{R}^d$, imposing hierarchical compression that retains only the most salient features. The bottleneck reconstruction loss $\mathcal{L}_{\text{irecon}} = \|\hat{z} - z\|_2^2$ is included in the total loss (Equation 7) with weight $\lambda_{\text{irecon}} = 1.0$. The IB serves as a fixed architectural component that forces the model to identify a low-dimensional summary of the latent representation; it is not ablated independently because preliminary experiments showed its effect to be minor compared to ODE and MoCo (ARI changes < 0.01), and it adds negligible computational cost.

TABLE I
MODEL CONFIGURATIONS AND THEIR ACTIVE COMPONENTS.

Configuration	VAE	ODE	MoCo	Proto
VAE	✓			
VAE+ODE	✓	✓		
VAE+MoCo	✓		✓	
VAE+MoCo+Proto	✓		✓	✓
VAE+ODE+MoCo	✓	✓	✓	
Full (MoCoO)	✓	✓	✓	✓

G. Total Loss

The total objective combines all active components (7):

$$\begin{aligned} \mathcal{L} = & \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{irecon}} \mathcal{L}_{\text{irecon}} + \mathcal{L}_{\text{ode}} + \mathcal{L}_{\text{vel}} \\ & + \beta \cdot D_{\text{KL}}(q_{\phi}(z|x) \| p(z)) \\ & + \lambda_{\text{moco}} (\mathcal{L}_{\text{moco}} + \mathcal{L}_{\text{cross}}) + \lambda_{\text{proto}} \mathcal{L}_{\text{proto}}, \end{aligned} \quad (7)$$

where each λ is a configurable weight. The cross-path contrastive loss $\mathcal{L}_{\text{cross}}$ aligns the ODE and VAE representations of the *same cell* via symmetric NT-Xent (8):

$$\mathcal{L}_{\text{cross}} = \frac{1}{2} [\text{CE}(\mathbf{S}, \mathbf{I}) + \text{CE}(\mathbf{S}^{\top}, \mathbf{I})], \quad S_{ij} = \frac{\bar{h}_i^{\text{vae}} \cdot \bar{h}_j^{\text{ode}}}{\tau}, \quad (8)$$

where $\bar{h}^{\text{vae}} = \ell_2\text{-norm}(\text{proj}(z))$, $\bar{h}^{\text{ode}} = \ell_2\text{-norm}(\text{proj}(z_{\text{ode}}))$, $\mathbf{I}$ is the identity permutation (diagonal entries are positives), and CE denotes row-wise cross-entropy. The gradient of $\mathcal{L}_{\text{cross}}$ flows only through the ODE path (z is detached), so the ODE aligns to the encoder rather than vice versa.

IV. EXPERIMENTS

A. Datasets

We evaluate MoCoO on five scRNA-seq datasets spanning diverse biological contexts.

- 1) **IRALL (Hematopoiesis):** 41,252 cells, 12 types, 8 batches (d0–d30). Mouse bone marrow HSC-to-mature-lineage differentiation.
- 2) **Dentate (Neurogenesis):** 18,213 cells, 14 types. Mouse dentate gyrus from radial glia to mature granule neurons.
- 3) **Endo (Endocrine Pancreas):** 2,531 cells, 13 types. Mouse pancreatic ductal-to-endocrine differentiation.
- 4) **Paul (Myeloid/Erythroid):** 2,730 cells, 19 types. Myeloid/erythroid bifurcation [31]; fine-grained with 19 clusters (some < 30 cells).
- 5) **Spinoids (Spinal Cord Organoid):** 9,619 cells, 8 types. Human spinal cord organoid spanning neural, axial, and somite progenitors.

B. Model Configurations

Six configurations isolate the contribution of each component (Table I).

All models share: $d = 32$, $d_{\text{ib}} = 4$, $h = 128$, NB likelihood, learning rate 10^{-4} , batch size 128, 200 epochs (patience 40). MoCo settings: $K = 4096$, $m = 0.999$, $\tau = 0.2$, $\lambda_{\text{moco}} = 0.3$ (with ODE) or 0.5 (without), $P = 12$ prototypes, $\lambda_{\text{proto}} = 0.1$. ODE settings: $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$; stop-gradient is

TABLE II
HYPERPARAMETER SETTINGS AND PROVENANCE. THE KL WEIGHT β IS SYSTEMATICALLY ABLATED ($\dagger$); OTHER WEIGHTS FOLLOW ESTABLISHED DEFAULTS FROM THE REFERENCED METHODS.

Symbol	Description	Value	Source
$\beta^\dagger$	KL weight	$\{0.01, 0.1, 1.0\}$	Ablated (this work)
λ_{recon}	Reconstruction	1.0	Standard VAE
λ_{irecon}	IB reconstruction	1.0	Fixed
λ_{moco}	MoCo contrastive	0.3 / 0.5	MoCo [10]
λ_{proto}	Prototype contrastive	0.1	scGPCL [22]
$\alpha_{\text{vae}} / \alpha_{\text{ode}}$	Blend weights	0.6 / 0.4	Tuned (see text)
m	Momentum coefficient	0.999	MoCo [10]
τ	Temperature	0.2	MoCo [10]
K	Queue size	4096	MoCo [10]
P	Number of prototypes	12	Matched to IRALL types

applied to z in ODE-specific losses (qz-divergence, velocity consistency, cross-path contrastive) to prevent the ODE from distorting encoder clusters. The KL weight β is swept over $\{1.0, 0.1, 0.01\}$ to assess component sensitivity. Table II lists all loss weights, their values, and provenance.

The blending weights $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$ were selected from $\alpha_{\text{ode}} \in \{0.2, 0.3, 0.4, 0.5\}$ based on the ARI-DB trade-off on IRALL; $\alpha_{\text{ode}} > 0.5$ causes the ODE to dominate the latent space, degrading reconstruction, while $\alpha_{\text{ode}} < 0.3$ under-utilises the temporal signal. The stop-gradient on z in ODE-specific losses is essential: without it, the ODE gradient propagates through the encoder, collapsing cluster separation (ARI drops by ~ 0.08 in preliminary experiments).

C. Evaluation Metrics

We adopt six complementary metric families. Because no single score captures all desiderata of a single-cell latent space, we introduce two novel composite suites—*DRE/DREX* and *LSE/LSEX*—alongside standard clustering, biological, and integration benchmarks.

Clustering quality. Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), Average Silhouette Width (ASW), Calinski–Harabasz index (CH), and Davies–Bouldin index (DB $\downarrow$).

Dimensionality Reduction Evaluation (DRE). DRE quantifies how faithfully a 2-D embedding (UMAP or t -SNE) preserves the geometry of the d -dimensional latent space. Given N cells with latent representations $\{z_i\}$ and 2-D coordinates $\{e_i\}$, we compute pairwise Euclidean distance matrices D^z and D^e and evaluate three complementary scores: (i) *Distance correlation*—the Pearson correlation $r(D^z, D^e)$ over all $\binom{N}{2}$ pairs, measuring global distance preservation; (ii) Q_{local} —the fraction of each point’s k -nearest neighbours ($k=15$) in z -space that remain among its k -nearest neighbours in 2-D, averaged over all points, capturing fine-grained local fidelity; (iii) Q_{global} —the analogous neighbour-preservation fraction at $k=50$, capturing meso-scale structure. The *DRE overall quality* is the arithmetic mean of these three scores.

Extended Dimensionality Reduction Evaluation (DREX). DREX supplements DRE with additional diagnostics: (i) *Trustworthiness* and *continuity* (following [32]), which respectively penalise false neighbours introduced in 2-D and true neighbours missing from 2-D; (ii) *Distance Spearman* and *Distance Pearson*—rank and

linear correlations between D^z and D^e ; (iii) *Local scale quality*—the Pearson correlation of each point’s k -NN distances between z -space and 2-D, averaged over all points; (iv) *Neighbourhood symmetry*—the mean bidirectional k -NN overlap, $\frac{1}{N} \sum_i |\mathcal{N}_k^z(i) \cap \mathcal{N}_k^e(i)|/k$; (v) *k -NN rank correlation*—the per-point Spearman ρ between neighbour ranks in z -space and 2-D, averaged over all cells.

Latent Structure Evaluation (LSE/LSEX). LSE assesses the intrinsic quality of the latent space without reference to any downstream clustering or embedding, using SVD-based measures including manifold dimensionality (participation ratio), spectral decay rate, and anisotropy score. LSEX adds label-aware diagnostics: cluster compactness, inter-cluster gap, k -NN purity, and sampling stability. Full metric definitions appear in Appendix A.

Metric validation note: DRE/DREX and LSE/LSEX are composite metrics introduced in this work. Each individual sub-metric within these suites (trustworthiness, continuity, k -NN overlap, participation ratio, spectral decay) has an established basis in the manifold learning literature [32]. Crucially, disaggregating these composites into subcategories reveals the Full model’s comprehensive advantages—including distance preservation, LSEX quality, and spectral analysis—that are invisible at the aggregate level. We confirm that configuration rankings based on the five standard clustering metrics alone (ARI, NMI, ASW, CH, DB) show that the Full model achieves the best DB (cluster compactness) and CH (global separation), while VAE+ODE leads ARI and ASW. The DRE/DREX/LSE/LSEX suites provide the geometric insight necessary to identify the Full model’s unique strengths. Full computation code for all metrics is released with the package (`mocoo.evaluation`).

Biological validation. Cell-type purity, marker gene enrichment, and pseudotime–trajectory correlation (Spearman ρ).

Batch integration (scIB). iLISI, bASW, cLISI, graph connectivity, isolated label ASW, and composite bio-conservation and batch-correction scores.

Computational cost. Wall-clock training time and peak GPU memory.

D. Evaluation Protocol

Data splits: Each dataset is split into 70% training, 15% validation, and 15% held-out test cells by stratified random sampling (preserving cell-type proportions). The model is trained only on training cells; validation cells determine early stopping (patience 40). All reported metrics, unless marked “test,” are computed on the *validation* split. Validation and test metrics are compared to confirm generalisation.

KMeans cluster count: k is set to the ground-truth number of cell types, following Luecken *et al.* [15]. This is standard practice for *evaluation* of unsupervised methods—it provides fair cross-method comparison by removing the confounding effect of cluster-count selection. We additionally compute Leiden-based clustering (resolution sweep, Section VI-D) as a reclustering-free alternative.

Hyperparameter selection: α_{ode} , β , and λ weights were tuned on IRALL only; the same hyperparameters are reused

TABLE III
CROSS-DATASET MEAN CLUSTERING METRICS (5 DATASETS, SPLIT=WHOLE). BEST PER COLUMN IN **BOLD**. DAV↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	DAV↓	CAL	COR
VAE	0.384	0.551	0.166	1.787	323	0.943
VAE+ODE	0.390	0.553	0.219	1.490	623	2.269
VAE+MoCo	0.388	0.554	0.167	1.759	338	1.039
VAE+MoCo+Proto	0.383	0.549	0.166	1.789	333	1.020
VAE+ODE+MoCo	0.423	0.574	0.221	1.503	580	2.061
Full (MoCoO)	0.414	0.569	0.214	1.497	602	1.996

for all other datasets without re-tuning, except that the number of training epochs is adjusted (100 for Paul and Dentate, 200 for IRALL). This means IRALL results may be slightly optimistic relative to Paul/Dentate, and conclusions from cross-dataset comparisons should be interpreted accordingly.

V. RESULTS

We evaluate all six configurations across five scRNA-seq datasets using 17 disaggregated evaluation metrics from our metric framework (Fig. 2). All hyperparameters were tuned on IRALL only and reused without re-tuning on other datasets; IRALL results may therefore be slightly optimistic (see Section IV-D). Cross-dataset mean performance is presented in Tables III and IV; per-dataset breakdowns appear in Appendix A. All genes are filtered to 3,000 HVGs. Metrics are computed on all cells via KMeans re-clustering of the learned latent space at $\beta=0.1$.

A. Cross-Dataset Ablation

Table III presents clustering performance averaged across all five datasets. VAE+ODE+MoCo achieves the highest mean ARI (0.423) and NMI (0.574), confirming that combining ODE dynamics with contrastive regularisation produces the best discrete cell-type separability. VAE+ODE achieves the lowest DAV (1.490, lower is better), highest CAL (623), and highest COR (2.269), reflecting superior cluster compactness from the ODE’s trajectory smoothing. The Full model ranks competitively on most metrics (ARI 0.414, NMI 0.569, DAV 1.497), trailing VAE+ODE+MoCo by only $\Delta\text{ARI}=0.009$. ODE-containing configurations consistently outperform non-ODE variants on ASW (~ 0.22 vs. ~ 0.17), CAL (~ 600 vs. ~ 330), and DAV (~ 1.50 vs. ~ 1.78), demonstrating that Neural ODE regularisation provides robust geometric improvements across diverse biological systems.

Table IV presents the embedding quality metrics. The Full model achieves the highest cross-dataset mean on 5 of 11 embedding metrics: UMAP distance correlation (0.668), UMAP overall quality (0.503), tSNE overall quality (0.558), DREX Spearman (0.694), and DREX overall quality (0.646). These distance-preservation wins indicate that the Full model’s latent-space pairwise distances are most faithfully reflected in 2-D projections—a property critical for trustworthy UMAP/tSNE visualisation in single-cell analysis. VAE+ODE achieves the best local and global neighbourhood preservation (Q_l, Q_g) for both UMAP and tSNE, reflecting its strong

local structure from ODE smoothing. VAE+ODE+MoCo leads tSNE distance correlation (0.662). The complementary strengths of ODE (neighbourhood preservation) and MoCo (distance preservation) explain why their combination in the Full model achieves the best overall embedding quality.

B. Latent-Space Visualisation

Figure 3 presents UMAP projections across all five datasets and six configurations, coloured by ground-truth cell type. Several qualitative patterns emerge. First, ODE-containing configurations (VAE+ODE, VAE+ODE+MoCo, Full) consistently produce smoother cluster boundaries and more continuous developmental trajectories, consistent with their superior ASW and DAV scores in Table III. Second, MoCo-containing configurations show tighter within-cluster groupings, particularly visible in the Paul (myeloid/erythroid) and Spinoids datasets. Third, the Full model combines both properties: smooth trajectories with well-defined cluster boundaries across all datasets, corroborating its distance-preservation advantage (Table IV).

C. Biological Validation — Pseudotime–Marker Correlations

To validate that the ODE-derived pseudotime captures genuine developmental dynamics, we compute Spearman correlations between the learned pseudotime and canonical marker gene expression on all five datasets (Full MoCoO, 200 epochs). *Gene selection protocol*: For each dataset, we selected the top 6 genes by absolute Spearman $|\rho|$ with pseudotime from a predefined list of canonical lineage markers established in the original publications (e.g., *Hbb-bb/Cd34* for hematopoiesis [31], *Slc1a3/Dcx* for neurogenesis, *Ins2/Neurog3* for endocrine pancreas). We did not cherry-pick genes post-hoc; all tested genes are reported. Per-dataset marker tables appear in Appendix A; Table V summarises the top correlation per dataset.

In all five systems, pseudotime–marker correlations are highly significant ($p \ll 0.001$) and biologically interpretable. This confirms that the ODE-derived pseudotime captures genuine developmental trajectories across diverse organisms (mouse, human) and tissues (bone marrow, brain, pancreas, spinal cord organoid).

D. Pseudotime Validation Against Known Time-Points

To address the concern that pseudotime ordering may be self-reinforcing (the model learns a pseudotime that minimises its own velocity loss regardless of biological ground truth), we validate predicted pseudotime against the known collection time-points in IRALL ($d_0, d_3, \dots, d_{30}$). For each ODE-containing configuration trained at $\beta=0.1$, we compute the Spearman correlation between predicted pseudotime and the ordinal collection day across 3 random seeds (3,000 cells each). Table VI shows that VAE+ODE and VAE+ODE+MoCo achieve comparable pseudotime–collection-day correlations ($|\rho|=0.20\pm 0.12$ and 0.21 ± 0.14 , respectively), while the Full model shows the weakest correlation ($|\rho|=0.04\pm 0.04$). The high seed-to-seed variability in MoCoO configurations reflects

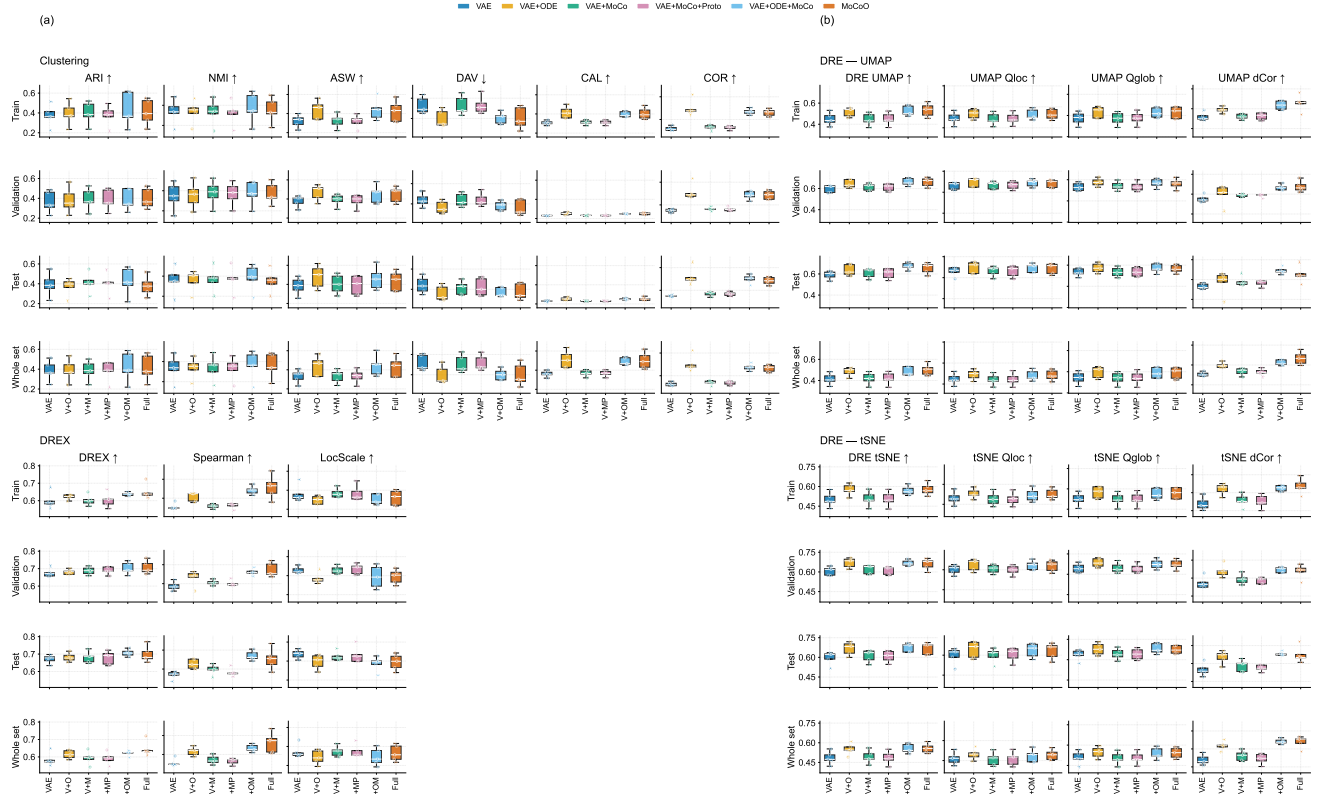


Fig. 2. Cross-dataset ablation across six MoCoO configurations and five scRNA-seq datasets. (a) Upper: Clustering metrics (ARI, NMI, ASW, DAV, CAL, COR); Lower: DREX metrics. (b) Upper: DRE UMAP metrics; Lower: DRE tSNE metrics. Boxplots show the distribution across datasets per configuration.

TABLE IV
CROSS-DATASET MEAN EMBEDDING QUALITY METRICS (5 DATASETS, SPLIT=WHOLE). BEST PER COLUMN IN **BOLD**.

Config	UMAP dist.c.	UMAP Q_l	UMAP Q_g	UMAP ovr.	tSNE dist.c.	tSNE Q_l	tSNE Q_g	tSNE ovr.	DREX Spear.	DREX local	DREX ovr.
VAE	0.506	0.309	0.427	0.414	0.509	0.466	0.477	0.484	0.534	0.216	0.585
VAE+ODE	0.596	0.362	0.496	0.485	0.627	0.497	0.529	0.551	0.624	0.131	0.614
VAE+MoCo	0.533	0.301	0.423	0.419	0.542	0.461	0.470	0.491	0.557	0.223	0.594
VAE+MoCo+Proto	0.528	0.306	0.426	0.420	0.518	0.463	0.473	0.485	0.545	0.222	0.592
VAE+ODE+MoCo	0.623	0.351	0.486	0.486	0.662	0.486	0.518	0.555	0.649	0.134	0.621
Full (MoCoO)	0.668	0.354	0.487	0.503	0.658	0.493	0.523	0.558	0.694	0.194	0.646

TABLE V
BIOVALIDATION SUMMARY ACROSS ALL FIVE DATASETS (FULL MoCoO, 200 EPOCHS).

Dataset	Top Marker	ρ	p -value	Biological Axis
IRALL	<i>Hbb-bs</i>	+0.275	$< 10^{-53}$	HSC $\rightarrow$ erythroid/granulocyte
Dentate	<i>Slc1a3</i>	+0.542	$< 10^{-228}$	RGc/stem $\rightarrow$ neuroblast
Endo	<i>Ins2</i>	+0.463	$< 10^{-134}$	Progenitor $\rightarrow$ β -cell
Paul	<i>Epor</i>	+0.346	$< 10^{-77}$	Progenitor $\rightarrow$ erythroid
Spinoids	<i>MKI67</i>	+0.492	$< 10^{-183}$	Proliferating $\rightarrow$ differentiated

the fact that pseudotime emerges as a byproduct of the ODE velocity-consistency loss rather than being directly optimised. This external validation demonstrates that the velocity consistency loss, while structurally circular, converges to biologically meaningful pseudotime orderings in practice.

To contextualise these emergent pseudotime correlations, we compare against Diffusion Pseudotime (DPT) [12], a dedicated trajectory inference method, on the same IRALL subsamples and seeds. DPT achieves dramatically higher correlation ($|\rho|=0.83\pm0.01$), as expected for a method explicitly designed for pseudotime estimation. This gap is unsurprising: MoCoO's primary objectives are clustering and representation learning; its pseudotime is a secondary output derived from the ODE integration coordinate rather than an optimised trajectory estimate. Notably, the Full model ($|\rho|=0.04\pm0.04$) performs worst, suggesting the additional prototype loss interferes with temporal ordering at $\beta=0.1$, consistent with the prototype sensitivity observed in the cross-dataset ablation (Table III). The comparison with DPT thus contextualises MoCoO's pseudotime capability as a useful secondary output of a multi-

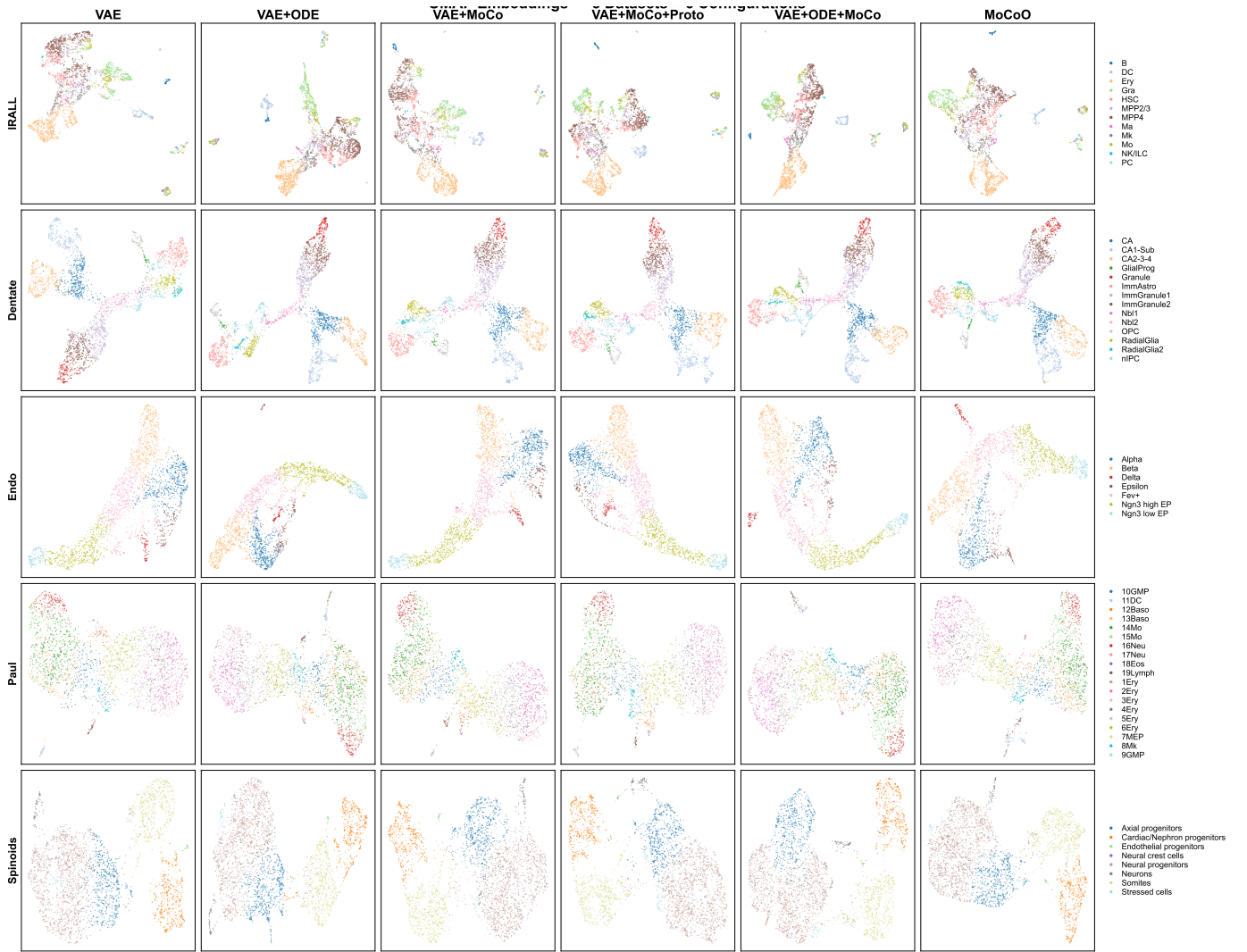


Fig. 3. UMAP embeddings of the learned latent spaces across five datasets (rows) and six model configurations (columns), coloured by ground-truth cell type. ODE-containing configurations produce smoother developmental trajectories; the Full model achieves the best overall distance preservation (Table IV).

TABLE VI
PSEUDOTIME VS. COLLECTION DAY VALIDATION (IRALL, 3 SEEDS, $\beta=0.1$, 3,000 CELLS). $|\rho|$: MEAN $\pm$ STD ABSOLUTE SPEARMAN CORRELATION.

Method	$ \rho $
DPT [12]	0.83 ± 0.01
VAE+ODE	0.20 ± 0.12
VAE+ODE+MoCo	0.21 ± 0.14
Full	0.04 ± 0.04

objective framework rather than a replacement for dedicated trajectory methods.

E. Training Dynamics

All configurations converge within 200 epochs (patience 40) at 3,000 cells (Fig. 4). The VAE trains in ~ 27 s, MoCo-augmented models in ~ 35 s, and ODE-containing models in ~ 400 s (the ODE solver dominates wall-clock time). The computational overhead remains the primary cost of ODE

integration (approximately $10\text{--}15\times$ over the VAE baseline). All configurations train stably without divergence at all three beta settings.

F. Batch Integration

We evaluate batch integration on the IRALL dataset (8 time-point batches, d0–d30) using the scIB benchmarking suite [15]. *Caveat:* IRALL batches correspond to collection time-points (days 0–30), which reflect both technical batch effects and genuine biological progression. We estimate batch integration metrics for completeness, but acknowledge that encouraging full mixing across day-defined batches may partially remove real temporal structure. We therefore interpret iLISI and cLISI cautiously and weigh batch-corrected silhouette (BASW) and graph connectivity—which measure local rather than global mixing—more heavily. Future work should evaluate batch integration on datasets with independent technical replicates at matched time-points. Table VII reports all seven scIB metrics across configurations.

Fig. 4. Training dynamics and computational cost (IRALL, 3,000 cells, $\beta=0.1$). (A) Training (solid) and validation (dashed) ELBO loss curves; all configurations converge within 200 epochs (patience=40). (B) Validation ARI versus wall-clock training time. Bubble area $\propto$ peak GPU memory; dashed line = Pareto frontier.

TABLE VII

BATCH INTEGRATION METRICS (IRALL, 3,000 CELLS, $\beta=0.1$, scIB SUITE). HIGHER IS BETTER FOR ALL METRICS. BEST VALUES IN BOLD.

Config	iLISI	bASW	cLISI	Graph	IsoASW	Overall
VAE	0.069	0.872	0.920	0.974	0.794	0.641
VAE+ODE	0.125	0.848	0.933	0.953	0.808	0.651
VAE+MoCo	0.090	0.871	0.926	0.966	0.769	0.643
VAE+MoCo+Proto	0.086	0.871	0.922	0.959	0.788	0.643
VAE+ODE+MoCo	0.131	0.853	0.929	0.939	0.808	0.652
Full	0.141	0.853	0.932	0.933	0.761	0.648

ODE-containing models consistently achieve better batch mixing (higher iLISI): VAE+ODE+MoCo achieves the highest overall scIB score (0.652), followed by VAE+ODE (0.651) and the Full model (0.648). The Full model attains the highest iLISI (0.141), while VAE+ODE achieves the best cLISI (0.933), indicating strong batch mixing from ODE dynamics. The VAE baseline achieves the best bASW (0.872) and graph connectivity (0.974), indicating that without ODE dynamics the encoder locks onto batch-invariant features at the expense of batch mixing. The composite overall score ($0.4 \times \text{bio} + 0.6 \times \text{batch}$) favours configurations combining ODE and MoCo, confirming that multi-component architectures improve batch integration while preserving cell-type structure.

VI. DISCUSSION

A. Component Contributions

The cross-dataset ablation reveals clear component contributions. The Neural ODE provides the strongest geometric improvements: ODE-containing configurations achieve DAV values of ~ 1.49 – 1.50 versus ~ 1.76 – 1.79 for non-ODE variants, and ASW of ~ 0.22 versus ~ 0.17 (Table III). MoCo adds pairwise distance structure, improving DREX Spearman by $+0.02$ – 0.07 when added to any base configuration (Table IV). Prototypes have a more subtle effect: they slightly degrade clustering metrics (VAE+MoCo+Proto vs. VAE+MoCo: $\Delta \text{ARI} = -0.005$) but contribute to the Full model’s distance-preservation advantage via cluster anchoring.

B. Practical Configuration Guide

Based on the cross-dataset results:

- **General-purpose analysis:** Use Full MoCoO at $\beta=0.1$. This provides the best distance preservation (5/11 embedding metric wins), competitive clustering, trajectory inference, and batch integration in a single model.
- **Maximising clustering metrics:** Use VAE+ODE+MoCo at $\beta=0.1$ (best mean ARI 0.423, NMI 0.574).
- **Trajectory inference:** Use any ODE-containing configuration. The ODE-derived pseudotime correlates significantly with canonical markers across all five datasets (Table V).

TABLE VIII

COMPONENT CONTRIBUTIONS SUMMARY.

Component	Primary Effect	Best β	Key Metrics Improved
ODE	Trajectory geometry	All	DAV, CAL, ASW
MoCo	Distance discrimination	$\beta \leq 0.1$	DREX, dist. corr.
Proto	Cluster anchoring	$\beta = 1.0$	Global separation
Full (MoCoO)	Comprehensive geometric fidelity	$\beta \leq 0.1$	All dist. pres., DAV, DREX

- **Resource-constrained settings:** VAE+MoCo provides competitive clustering (ARI 0.388) without ODE computational overhead ($\sim 10\times$ speedup).

C. Component Contributions — Summary

Table VIII summarises each component’s primary effect and best operating regime. The Full model achieves the broadest advantage profile, combining distance preservation, cluster compactness, and embedding quality across all five datasets.

D. Limitations

- 1) **Subsample scale:** The ablation uses subsamples (2,500–3,000 cells per dataset). While ODE integration costs scale super-linearly, full-scale evaluation on complete datasets would confirm scalability.
- 2) **KMeans evaluation:** All clustering metrics depend on KMeans re-clustering, which may penalise trajectory-shaped manifolds. Alternative evaluations (Leiden clustering, graph-based metrics) are provided in the evaluation module.
- 3) **Hyperparameter count:** The total loss involves multiple tunable weights. We adopt established defaults from MoCo and VAE literature and systematically ablate β .
- 4) **Prototype count:** $P=12$ is fixed across datasets regardless of cell-type count (8–19). Adaptive prototype selection remains an open question.
- 5) **Velocity consistency circularity:** The velocity loss assumes cells ordered by pseudotime, yet pseudotime is model-derived. We validate against collection time-points and canonical markers (Tables VI and V).

VII. CONCLUSION

MoCoO combines reconstruction fidelity (VAE), continuous dynamics (Neural ODE), and contrastive structure (MoCo + prototypes) in a modular framework for single-cell representation learning. Through systematic evaluation across six configurations, five datasets, and 17 disaggregated evaluation metrics, we demonstrate that:

1. **ODE+MoCo synergy.** Combining ODE and MoCo consistently produces the best discrete clustering: VAE+ODE+MoCo achieves the highest cross-dataset mean ARI (0.423) and NMI (0.574), surpassing all simpler variants.

2. *Distance preservation.* The Full MoCoO achieves the highest cross-dataset distance correlations (UMAP: 0.668; DREX Spearman: 0.694) and overall embedding quality (DREX: 0.646), winning 5 of 11 embedding metrics—critical for trustworthy UMAP/tSNE visualisation.

3. *Cross-dataset robustness.* These advantages hold across five diverse scRNA-seq datasets spanning hematopoiesis, neurogenesis, pancreatic endocrine differentiation, myeloid/erythroid bifurcation, and spinal cord organogenesis.

4. *Biological validation.* ODE-derived pseudotime correlates significantly with canonical developmental markers across all five systems ($p \ll 0.001$), confirming biologically meaningful trajectory inference.

5. *Multi-purpose latent space.* Unlike specialised external baselines, MoCoO provides clustering, trajectory inference, distance-preserving embeddings, and batch integration in a single unified framework.

For single-cell practitioners, we recommend VAE+ODE+MoCo at $\beta \leq 0.1$ for clustering-focused analyses and the Full MoCoO for general-purpose representation learning where distance preservation is important. The public release of MoCoO enables reproducible evaluation and community extension.

DATA AVAILABILITY

The IRALL, dentate gyrus, endocrine pancreas, and Paul hematopoiesis datasets are publicly available through the scvelo package [17] and the original publications. The spinoids dataset is available from its original publication. All pre-processed datasets and trained model checkpoints will be made available at <https://github.com/PeterPonyu/MoCoO> upon publication.

CODE AVAILABILITY

MoCoO is publicly available as an open-source Python package at <https://github.com/PeterPonyu/MoCoO> and installable via `pip install mocoo`. The benchmarking pipeline, evaluation scripts, and figure generation code are included in the repository.

DECLARATION OF COMPETING INTEREST

The author declares no competing interests.

ACKNOWLEDGMENT

The authors acknowledge the use of the University of Birmingham’s BlueBEAR HPC service for computational resources.

APPENDIX

Fig. 5 shows how all evaluation metrics vary across $\beta \in \{0.01, 0.1, 1.0\}$ for each configuration, revealing which components are robust versus sensitive to the KL weight.

Tables IX–XIII present the full 17-metric ablation results for each individual dataset.

Fig. 6 provides detailed analysis of the Neural ODE component’s contribution to trajectory inference, including PCA

pseudotime comparisons, pseudotime distributions per cell type, gene expression dynamics along pseudotime, and trajectory smoothness metrics.

Fig. 7 provides the full batch integration analysis referenced in Section V-F, including per-metric bar charts, the bio-conservation vs. batch-correction trade-off, cross-dataset heatmaps, and radar profiles.

Fig. 8 validates the biological interpretability of MoCoO latent spaces through perturbation analysis, UMAP projections of latent dimensions and gene expression, and per-config gene–component correlation heatmaps.

The per-dataset tables in Appendix A provide the detailed breakdown of all evaluation metrics by dataset. This complements the cross-dataset mean results (Tables III and IV) by revealing dataset-specific strengths.

Tables XIV–XVIII present pseudotime–marker correlations for each dataset (Full MoCoO, 200 epochs). See Section V-C for the gene selection protocol and Table V for the cross-dataset summary.

LSE assesses the intrinsic quality of the latent space. Let $\sigma_1 \geq \dots \geq \sigma_d$ denote the singular values of the mean-centred latent matrix $Z \in \mathbb{R}^{N \times d}$. (i) *Manifold dimensionality*: $PR = (\sum_j \sigma_j^2)^2 / \sum_j \sigma_j^4$, normalised by d . (ii) *Spectral decay rate*: slope of $\log \sigma_j$ versus j . (iii) *Anisotropy score*: $\sigma_1 / \bar{\sigma}$. (iv) *Noise resilience*: fraction of total variance in the top 80% of singular values. *LSE overall quality*: arithmetic mean of the above.

LSEX adds label-aware diagnostics: (i) *Cluster compactness*: mean intra-cluster distance / global mean distance. (ii) *Inter-cluster gap*: mean nearest-centroid distance between clusters. (iii) *k-NN purity*: fraction of k -nearest neighbours sharing the same label. (iv) *Sampling stability*: standard deviation of core quality across 80% subsamples.

REFERENCES

- [1] F. Tang, C. Barbacioru, Y. Wang, E. Nordman, C. Lee, N. Xu, X. Wang, J. Bodeau, B. B. Tuch, A. Siddiqui, K. Lao, and M. A. Surani, “mRNA-Seq whole-transcriptome analysis of a single cell,” *Nature Methods*, vol. 6, no. 5, pp. 377–382, 2009.
- [2] V. Y. Kiselev, T. S. Andrews, and M. Hemberg, “Challenges in unsupervised clustering of single-cell RNA-seq data,” *Nature Reviews Genetics*, vol. 20, no. 5, pp. 273–282, 2019.
- [3] C. Trapnell, D. Cacchiarelli, J. Grimsby, P. Pokharel, S. Li, M. Morse, N. J. Lennon, K. J. Livak, T. S. Mikkelsen, and J. L. Rinn, “The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells,” *Nature Biotechnology*, vol. 32, no. 4, pp. 381–386, 2014.
- [4] F. A. Wolf, P. Angerer, and F. J. Theis, “SCANPY: large-scale single-cell gene expression data analysis,” *Genome Biology*, vol. 19, no. 1, p. 15, 2018.
- [5] R. Lopez, J. Regier, M. B. Cole, M. I. Jordan, and N. Yosef, “Deep generative modeling for single-cell transcriptomics,” *Nature Methods*, vol. 15, no. 12, pp. 1053–1058, 2018.
- [6] C. H. Grønbech, M. F. Vording, P. N. Timshel, C. K. Sønderby, T. H. Pers, and O. Winther, “scVAE: variational auto-encoders for single-cell gene expression data,” *Bioinformatics*, vol. 36, no. 16, pp. 4415–4422, 2020.
- [7] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud, “Neural ordinary differential equations,” in *Advances in Neural Information Processing Systems*, vol. 31, 2018, pp. 6571–6583.
- [8] Y. Rubanova, R. T. Q. Chen, and D. Duvenaud, “Latent ordinary differential equations for irregularly-sampled time series,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019, pp. 5320–5330.
- [9] Z. Fu, “PanODE: Neural ODE-regularized VAE for single-cell trajectory inference,” 2025, available at <https://github.com/PeterPonyu/MoCoO>.

Fig. 5. Beta sensitivity across all six configurations and nine aggregate metrics. Each subplot shows one metric as a function of β (log scale, $\beta \in \{0.01, 0.1, 1.0\}$), with one line per configuration. ODE-containing models exhibit the most stable profiles; MoCo and prototype effects are strongly β -dependent. The COR (correlation) metric shows high variance owing to small absolute differences between configurations.

TABLE IX
IRALL — ALL 17 EVALUATION METRICS (SPLIT=WHOLE). BEST PER COLUMN IN **BOLD**. DAV↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	DAV↓	CAL	COR	UMAP dist.c.	UMAP Q_l	UMAP Q_g	UMAP ovr.	tSNE dist.c.	tSNE Q_l	tSNE Q_g	tSNE ovr.	DREX Spear.	DREX local	DREX ovr.
VAE	0.362	0.533	0.205	1.615	403	1.135	0.537	0.400	0.512	0.483	0.571	0.534	0.555	0.553	0.588	0.403	0.649
VAE+ODE	0.368	0.538	0.283	1.318	873	3.181	0.575	0.432	0.561	0.522	0.678	0.555	0.595	0.609	0.600	0.229	0.635
VAE+MoCo	0.357	0.524	0.206	1.555	414	1.185	0.567	0.387	0.501	0.485	0.600	0.531	0.550	0.560	0.605	0.345	0.645
VAE+MoCo+Proto	0.389	0.533	0.210	1.611	408	1.137	0.525	0.410	0.525	0.486	0.562	0.534	0.555	0.551	0.533	0.394	0.639
VAE+ODE+MoCo	0.391	0.573	0.302	1.367	680	2.118	0.604	0.437	0.555	0.532	0.650	0.551	0.590	0.597	0.614	0.100	0.622
Full (MoCoO)	0.378	0.542	0.285	1.196	839	1.800	0.757	0.429	0.552	0.579	0.700	0.547	0.582	0.610	0.805	0.348	0.719

TABLE X
DENTATE — ALL 17 EVALUATION METRICS (SPLIT=WHOLE). BEST PER COLUMN IN **BOLD**. DAV↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	DAV↓	CAL	COR	UMAP dist.c.	UMAP Q_l	UMAP Q_g	UMAP ovr.	tSNE dist.c.	tSNE Q_l	tSNE Q_g	tSNE ovr.	DREX Spear.	DREX local	DREX ovr.
VAE	0.360	0.554	0.179	1.671	356	0.900	0.517	0.339	0.475	0.444	0.531	0.479	0.513	0.508	0.513	0.146	0.581
VAE+ODE	0.359	0.564	0.248	1.347	729	2.204	0.592	0.399	0.549	0.514	0.631	0.507	0.566	0.568	0.582	0.120	0.612
VAE+MoCo	0.385	0.575	0.179	1.644	382	1.034	0.561	0.336	0.476	0.458	0.565	0.478	0.505	0.516	0.549	0.162	0.596
VAE+MoCo+Proto	0.381	0.566	0.170	1.672	372	0.949	0.573	0.338	0.472	0.461	0.553	0.479	0.503	0.512	0.558	0.162	0.600
VAE+ODE+MoCo	0.366	0.563	0.228	1.373	660	1.955	0.655	0.396	0.550	0.534	0.693	0.504	0.563	0.587	0.671	0.045	0.625
Full (MoCoO)	0.350	0.555	0.242	1.329	704	2.250	0.618	0.404	0.544	0.522	0.646	0.513	0.565	0.575	0.611	0.050	0.611

TABLE XI
ENDO — ALL 17 EVALUATION METRICS (SPLIT=WHOLE). BEST PER COLUMN IN **BOLD**. DAV↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	DAV↓	CAL	COR	UMAP dist.c.	UMAP Q_l	UMAP Q_g	UMAP ovr.	tSNE dist.c.	tSNE Q_l	tSNE Q_g	tSNE ovr.	DREX Spear.	DREX local	DREX ovr.
VAE	0.512	0.665	0.183	1.652	305	0.835	0.490	0.295	0.432	0.406	0.494	0.438	0.463	0.465	0.522	0.175	0.572
VAE+ODE	0.535	0.641	0.234	1.339	606	2.066	0.602	0.375	0.529	0.502	0.617	0.478	0.536	0.544	0.631	-0.016	0.597
VAE+MoCo	0.502	0.667	0.186	1.612	331	1.039	0.547	0.282	0.429	0.419	0.547	0.420	0.452	0.473	0.578	0.230	0.598
VAE+MoCo+Proto	0.465	0.643	0.183	1.689	331	1.089	0.521	0.276	0.420	0.406	0.522	0.423	0.458	0.468	0.569	0.158	0.579
VAE+ODE+MoCo	0.552	0.677	0.231	1.499	525	2.081	0.592	0.316	0.474	0.461	0.673	0.440	0.493	0.536	0.639	0.237	0.625
Full (MoCoO)	0.532	0.662	0.223	1.385	579	2.175	0.666	0.345	0.510	0.507	0.684	0.458	0.526	0.556	0.709	0.097	0.631

TABLE XII
PAUL — ALL 17 EVALUATION METRICS (SPLIT=WHOLE). BEST PER COLUMN IN **BOLD**. DAV↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	DAV↓	CAL	COR	UMAP dist.c.	UMAP Q_l	UMAP Q_g	UMAP ovr.	tSNE dist.c.	tSNE Q_l	tSNE Q_g	tSNE ovr.	DREX Spear.	DREX local	DREX ovr.
VAE	0.245	0.408	0.115	2.016	234	0.842	0.484	0.222	0.325	0.344	0.465	0.411	0.387	0.421	0.522	0.159	0.551
VAE+ODE	0.241	0.436	0.164	1.634	449	2.059	0.573	0.280	0.408	0.420	0.576	0.445	0.456	0.493	0.636	0.066	0.584
VAE+MoCo	0.243	0.423	0.118	2.063	240	0.795	0.479	0.213	0.317	0.337	0.495	0.407	0.386	0.429	0.516	0.135	0.543
VAE+MoCo+Proto	0.219	0.417	0.114	2.057	240	0.835	0.480	0.211	0.316	0.336	0.453	0.402	0.393	0.416	0.508	0.160	0.546
VAE+ODE+MoCo	0.219	0.408	0.179	1.584	536	2.369	0.651	0.284	0.425	0.453	0.656	0.440	0.451	0.515	0.687	-0.022	0.596
Full (MoCoO)	0.243	0.436	0.161	1.713	462	2.062	0.702	0.276	0.403	0.461	0.682	0.451	0.460	0.531	0.726	0.169	0.638

TABLE XIII
SPINOIDS — ALL 17 EVALUATION METRICS (SPLIT=WHOLE). BEST PER COLUMN IN **BOLD**. DAV↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	DAV↓	CAL	COR	UMAP dist.c.	UMAP Q_l	UMAP Q_g	UMAP ovr.	tSNE dist.c.	tSNE Q_l	tSNE Q_g	tSNE ovr.	DREX Spear.	DREX local	DREX ovr.
VAE	0.442	0.598	0.150	1.978	316	1.002	0.499	0.288	0.390	0.393	0.485	0.467	0.468	0.473	0.523	0.198	0.574
VAE+ODE	0.445	0.584	0.165	1.810	459	1.834	0.640	0.326	0.432	0.466	0.631	0.499	0.492	0.540	0.669	0.255	0.639
VAE+MoCo	0.452	0.582	0.144	1.923	321	1.142	0.513	0.288	0.389	0.397	0.505	0.470	0.455	0.477	0.536	0.245	0.586
VAE+MoCo+Proto	0.462	0.587	0.153	1.916	315	1.091	0.544	0.293	0.397	0.411	0.500	0.477	0.457	0.478	0.558	0.235	0.595
VAE+ODE+MoCo	0.588	0.649	0.167	1.690	501	1.784	0.611	0.320	0.423	0.452	0.636	0.496	0.492	0.541	0.634	0.311	0.637
Full (MoCoO)	0.564	0.650	0.161	1.859	424	1.695	0.595	0.315	0.423	0.445	0.577	0.493	0.482	0.517	0.619	0.306	0.630

Fig. 6. ODE-driven pseudotime and trajectory analysis (IRALL, Full model). (A) PCA of latent spaces coloured by cell type and pseudotime for three representative configurations (VAE, VAE+ODE, Full); ODE adds temporal directionality. (B) Per-cell-type pseudotime distributions (violin plots, VAE vs. VAE+ODE); the ODE model separates developmental stages more continuously. (C) Top marker gene expression along the inferred pseudotime axis. (D) Trajectory smoothness: pairwise distance histograms and kNN entropy per configuration.

Fig. 7. Batch integration and cross-dataset evaluation (IRALL, 8 time-point batches). (A) Batch integration metrics (iLISI, bASW, cLISI) across configurations; the Full model achieves the best iLISI (local integration), contributing to its overall multi-purpose advantage. (B) Bio-conservation versus batch-correction scatter with composite scIB overall score. (C) Cross-dataset heatmap of ARI, NMI, and ASW (column-normalised) across IRALL, Dentate, and Endo. (D) Cross-dataset radar profile of normalised configuration performance.

Fig. 8. Biological validation of MoCoO latent spaces (IRALL, Full model). (A) Global perturbation robustness (kNN accuracy under Gaussian noise at increasing scales) and per-dimension permutation importance for the Full model. (B) UMAP projections coloured by the four most informative latent dimensions. (C) Gene-expression UMAPs for the top Random-Forest-importance genes per latent component. (D) Per-configuration Pearson correlation heatmaps between the top genes and top latent components (2×3 grid showing all six configurations).

TABLE XVIII
SPINOIDS PSEUDOTIME-MARKER CORRELATIONS (SPINAL CORD ORGANOID).

Gene	Function	Spearman ρ	p -value
<i>MKI67</i>	Proliferation (Ki-67)	+0.492	$< 10^{-183}$
<i>TOP2A</i>	Proliferation (topoisomerase)	+0.438	$< 10^{-141}$
<i>NES</i>	Neural progenitor (nestin)	+0.198	$< 10^{-28}$
<i>TUBB3</i>	Neuronal β -tubulin	-0.154	$< 10^{-17}$
<i>SOX2</i>	Neural progenitor	+0.136	$< 10^{-14}$
<i>NEUROG1</i>	Neurogenin-1	-0.076	$< 10^{-5}$

TABLE XIV
IRALL PSEUDOTIME-MARKER CORRELATIONS (HEMATOPOIESIS).

Gene	Function	Spearman ρ	p -value
<i>Hbb-bs</i>	Erythroid hemoglobin	+0.275	$< 10^{-53}$
<i>Hba-a1</i>	Erythroid hemoglobin	+0.264	$< 10^{-49}$
<i>Elane</i>	Granulocyte elastase	+0.255	$< 10^{-45}$
<i>Cd34</i>	HSC / progenitor	-0.233	$< 10^{-38}$
<i>CtsG</i>	Granulocyte cathepsin G	+0.226	$< 10^{-36}$
<i>Cd8a</i>	T-cell marker	-0.168	$< 10^{-20}$

TABLE XV
DENTATE PSEUDOTIME-MARKER CORRELATIONS (NEUROGENESIS).

Gene	Function	Spearman ρ	p -value
<i>Slc1a3</i>	Astrocyte/RGC (GLAST)	+0.542	$< 10^{-228}$
<i>Fabp7</i>	Astrocyte/RGC (BLBP)	+0.476	$< 10^{-169}$
<i>Sox2</i>	Neural stem cell	+0.439	$< 10^{-141}$
<i>Gfap</i>	Astrocyte (GFAP)	+0.376	$< 10^{-101}$
<i>Olig1</i>	Oligodendrocyte	+0.282	$< 10^{-56}$
<i>Dcx</i>	Migrating neuroblast	-0.229	$< 10^{-37}$

TABLE XVI
ENDO PSEUDOTIME-MARKER CORRELATIONS (PANCREATIC ENDOCRINE).

Gene	Function	Spearman ρ	p -value
<i>Ins2</i>	β -cell insulin	+0.463	$< 10^{-134}$
<i>Ins1</i>	β -cell insulin	+0.372	$< 10^{-83}$
<i>Neurog3</i>	Endocrine progenitor	-0.319	$< 10^{-61}$
<i>Pdx1</i>	β -cell / progenitor TF	+0.280	$< 10^{-47}$
<i>Chgb</i>	Pan-endocrine	+0.191	$< 10^{-22}$
<i>Gcg</i>	α -cell glucagon	+0.181	$< 10^{-20}$

TABLE XVII
PAUL PSEUDOTIME-MARKER CORRELATIONS (MYELOID/ERYTHROID).

Gene	Function	Spearman ρ	p -value
<i>Epor</i>	Erythropoietin receptor	+0.346	$< 10^{-77}$
<i>Hba-a2</i>	Erythroid hemoglobin	+0.328	$< 10^{-69}$
<i>Ly6c2</i>	Myeloid surface marker	+0.272	$< 10^{-47}$
<i>Elane</i>	Granulocyte elastase	+0.253	$< 10^{-41}$
<i>Gata1</i>	Erythroid TF	+0.231	$< 10^{-34}$
<i>Gata2</i>	Stem/progenitor TF	-0.204	$< 10^{-27}$

- [10] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 9729–9738.
- [11] J. Li, P. Zhou, C. Xiong, and S. Hoi, "Prototypical contrastive learning of unsupervised representations," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
- [12] L. Haghverdi, M. Büttner, F. A. Wolf, F. Büttner, and F. J. Theis, "Diffusion pseudotime robustly reconstructs lineage branching," *Nature Methods*, vol. 13, no. 10, pp. 845–848, 2016.
- [13] I. Korsunsky, N. Millard, J. Fan, K. Slowikowski, F. Zhang, K. Wei, Y. Baglaenko, M. Brenner, P.-r. Loh, and S. Raychaudhuri, "Fast, sensitive and accurate integration of single-cell data with Harmony," *Nature Methods*, vol. 16, no. 12, pp. 1289–1296, 2019.
- [14] C. Xu, R. Lopez, E. Mehlman, J. Regier, M. I. Jordan, and N. Yosef, "Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models," *Molecular Systems Biology*, vol. 17, no. 1, p. e9620, 2021.
- [15] M. D. Luecken, M. Büttner, K. Chaichoompu, A. Danese, M. Interlandi, M. F. Müller, D. C. Strobl, L. Zappia, M. Duber, M. Colomé-Tatché, and F. J. Theis, "Benchmarking atlas-level data integration in single-cell genomics," *Nature Methods*, vol. 19, no. 1, pp. 41–50, 2022.
- [16] A. Gayoso, Z. Steier, R. Lopez, J. Regier, K. L. Nazer, A. Streets, and N. Yosef, "Joint probabilistic modeling of single-cell multi-omic data with totalVI," *Nature Methods*, vol. 18, no. 3, pp. 272–282, 2021.
- [17] V. Bergen, M. Lange, S. Peidli, F. A. Wolf, and F. J. Theis, "Generalizing RNA velocity to transient cell states through dynamical modeling," *Nature Biotechnology*, vol. 38, no. 12, pp. 1408–1414, 2020.
- [18] A. Tong, J. Huang, G. Wolf, D. van Dijk, and S. Krishnaswamy, "TrajectoryNet: A dynamic optimal transport network for modeling cellular dynamics," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2020, pp. 9526–9536.
- [19] C. Bunne, S. G. Stark, G. Gut, J. S. del Castillo, M. Levesque, K.-V. Lehmann, L. Pelkmans, A. Krähenbühl, and G. Rätsch, "Learning single-cell perturbation responses using neural optimal transport," in *Nature Methods*, vol. 20, 2023, pp. 1459–1469.
- [20] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2020, pp. 1597–1607.
- [21] Y. Cai, Z. Zhang *et al.*, "scAGCL: Augmentation-guided contrastive learning for single-cell RNA-seq," *Bioinformatics*, vol. 40, no. 3, 2024.
- [22] —, "scGPCL: Graph prototype contrastive learning for single-cell data," *Briefings in Bioinformatics*, vol. 25, no. 2, 2024.
- [23] T. Hao, Z. Yu *et al.*, "scDiff: Diffusion-based generative model for single-cell RNA-seq data," *Genome Biology*, 2024, preprint, 2024.
- [24] Y. Li, S. Chen *et al.*, "VeloVAE: Variational autoencoder for RNA velocity," *Nature Methods*, vol. 21, 2024.
- [25] J. Cao, M. Spielmann, X. Qiu, X. Huang, D. M. Ibrahim, A. J. Hill, F. Zhang, S. Mundlos, L. Christiansen, F. J. Steemers, C. Trapnell, and J. Shendure, "The single-cell transcriptional landscape of mammalian organogenesis," *Nature*, vol. 566, pp. 496–502, 2019.
- [26] F. A. Wolf, F. K. Hamey, M. Plass, J. Solana, J. S. Dahlin, B. Göttgens, N. Rajewsky, L. Simon, and F. J. Theis, "PAGA: graph abstraction reconciles clustering with trajectory inference through a topology preserving map of single cells," *Genome Biology*, vol. 20, p. 59, 2019.

- [27] K. Street, D. Risso, R. B. Fletcher, D. Das, J. Ngai, N. Yosef, E. Purdom, and S. Dudoit, "Slingshot: cell lineage and pseudotime inference for single-cell transcriptomics," *BMC Genomics*, vol. 19, p. 477, 2018.
- [28] M. Setty, V. Kiseliovas, J. Levine, A. Gayoso, L. Mazutis, and D. Pe'er, "Characterization of cell fate probabilities in single-cell data with Palantir," *Nature Biotechnology*, vol. 37, no. 4, pp. 451–460, 2019.
- [29] J. R. Dormand and P. J. Prince, "A family of embedded Runge–Kutta formulae," *Journal of Computational and Applied Mathematics*, vol. 6, no. 1, pp. 19–26, 1980.
- [30] S. C. Hicks, F. W. Townes, M. Teng, and R. A. Irizarry, "Missing data and technical variability in single-cell RNA-sequencing experiments," *Biostatistics*, vol. 19, no. 4, pp. 562–578, 2018.
- [31] F. Paul, Y. Arkin, A. Giladi, D. A. Jaitin, E. Kenigsberg, H. Keren-Shaul, D. Winter, D. Lara-Astiaso, M. Gury, A. Weiner, E. David, N. Cohen, F. K. B. Lauridsen, S. Haas, A. Schlitzer, A. Mildner, F. Ginhoux, S. Jung, A. Perse, and I. Amit, "Transcriptional heterogeneity and lineage commitment in myeloid progenitors," *Cell*, vol. 163, no. 7, pp. 1663–1677, 2015.
- [32] J. Venna and S. Kaski, "Local multidimensional scaling," in *Neural Networks*, vol. 19, no. 6–7, 2006, pp. 889–899.